

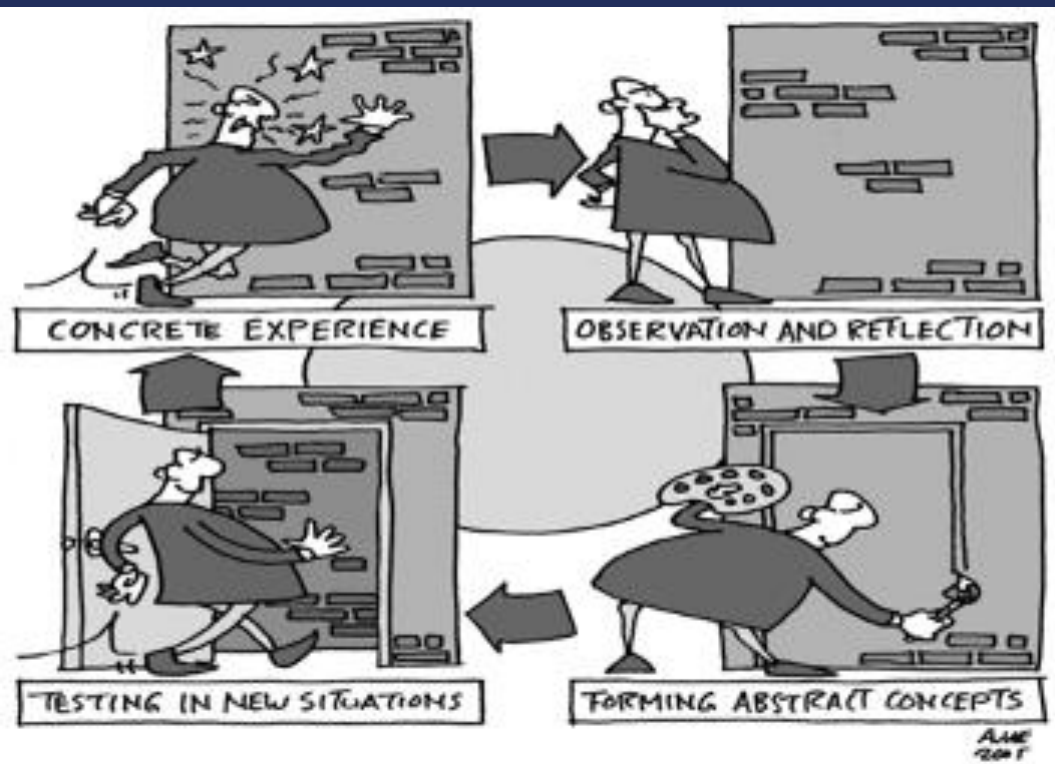


NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Faculty of Computing & Informatics

Action Research (AR) & Design-based Research

Towards relevant SW development research



Heike
Winschiers-Theophilus

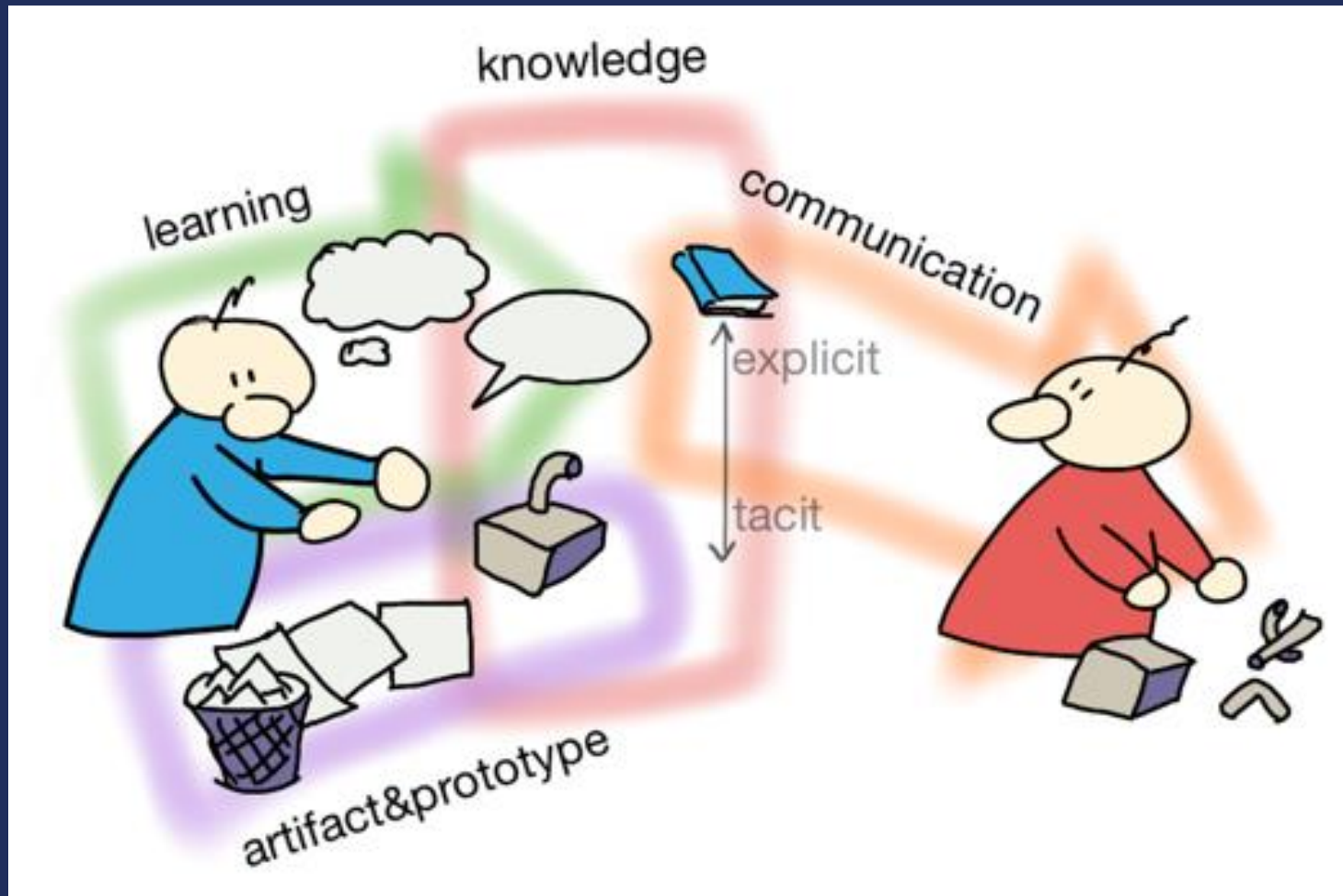
Terminology

	Working Definition	Key Associated Terms
Doing Research	Work done with the intention to produce knowledge for use by others	Question, hypothesis, theory, investigation, interpretation, generalization, validation, discovery
Doing Design	Work done with the intention to produce a feasible solution to improve a given situation	Idea and concept generation, synthesis, development, integration, discovery, prototyping, invention, implementation, realization
Artifact	Object (often material) created during a design process	Sketch, blueprint, brief, specifications, vision, proposal, recommendation, business plan
Prototype	Artifact used in research that can realize the (inter)action that is studied	Implementation, realization, test , exploration, solution, proof of concept, construction
Knowledge	Understanding about the world that can be communicated to others	Theory, book, publication, expertise
Design Practice	The ways in which design professionals conduct their work	Brief, contract, client, stakeholder, studio
Experiment	Specific: a piece of controlled, hypothesis-testing research; General: an explorative confrontation with real-world situations	Specific: hypothesis, statistics, (in)dependent variables; General: trial, intervention, exploration



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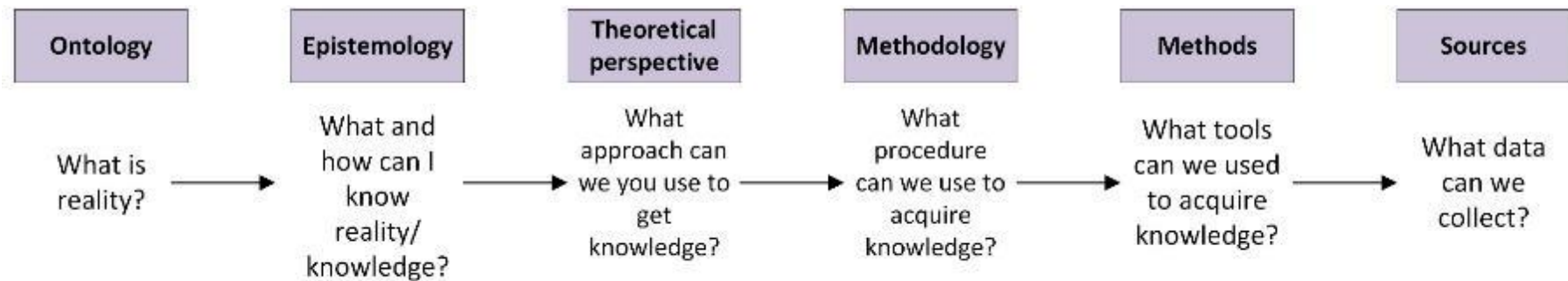
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Paradigm	Ontology <i>What is reality?</i>	Epistemology <i>How can I know reality?</i>	Theoretical Perspective <i>Which approach do you use to know something?</i>	Methodology <i>How do you go about finding out?</i>	Method <i>What techniques do you use to find out?</i>
Positivism	There is a single reality or truth (more realist).	Reality can be measured and hence the focus is on reliable and valid tools to obtain that.	Positivism Post-positivism	Experimental research Survey research	Usually quantitative, could include: Sampling Measurement and scaling Statistical analysis Questionnaire Focus group Interview
Constructivist / Interpretive	There is no single reality or truth. Reality is created by individuals in groups (less realist).	Therefore, reality needs to be interpreted. It is used to discover the underlying meaning of events and activities.	Interpretivism (reality needs to be interpreted) • Phenomenology • Symbolic interactionism • Hermeneutics Critical Inquiry Feminism	Ethnography Grounded Theory Phenomenological research Heuristic inquiry Action Research Discourse Analysis Feminist research etc	Usually qualitative, could include: Qualitative interview Observation Participant Non participant Case study Life history Narrative Theme identification etc
Pragmatism	Reality is constantly renegotiated, debated, interpreted in light of its usefulness in new unpredictable situations.	The best method is one that solves problems. Finding out is the means, change is the underlying aim.	Deweyan pragmatism <i>Research through design</i>	Mixed methods Design-based research Action research	Combination of any of the above and more, such as data mining expert review, usability testing, physical prototype
Subjectivism	Reality is what we perceive to be real	All knowledge is purely a matter of perspective.	Postmodernism Structuralism Post-structuralism	Discourse theory Archaeology Genealogy Deconstruction etc.	Autoethnography Semiotics Literary analysis Pastiche Intertextuality etc.
Critical	Realities are socially constructed entities that are under constant internal influence.	Reality and knowledge is both socially constructed and influenced by power relations from within society	Marxism Queer theory feminism	critical discourse analysis, critical ethnography action research ideology critique	Ideological review Civil actions open-ended interviews, focus groups, open-ended questionnaires, open-ended observations, and journals.

AR and Design-based research a Pragmatic Approach



Adapted from Hay (2002) pg. 64 and Crotty (1998)

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Value-laden/
subjective

Dependability
co-constructed

Learning
through
change

Open-ended/
iterative

Triangulation

Action Research

Motivations

- To make academic research relevant, researchers should try out their theories with practitioners in real situations and real organizations
- The emphasis is more on what practitioners do than on what they say they do

Key Assumptions

- Social settings cannot be reduced for study, and
- Actions brings understanding

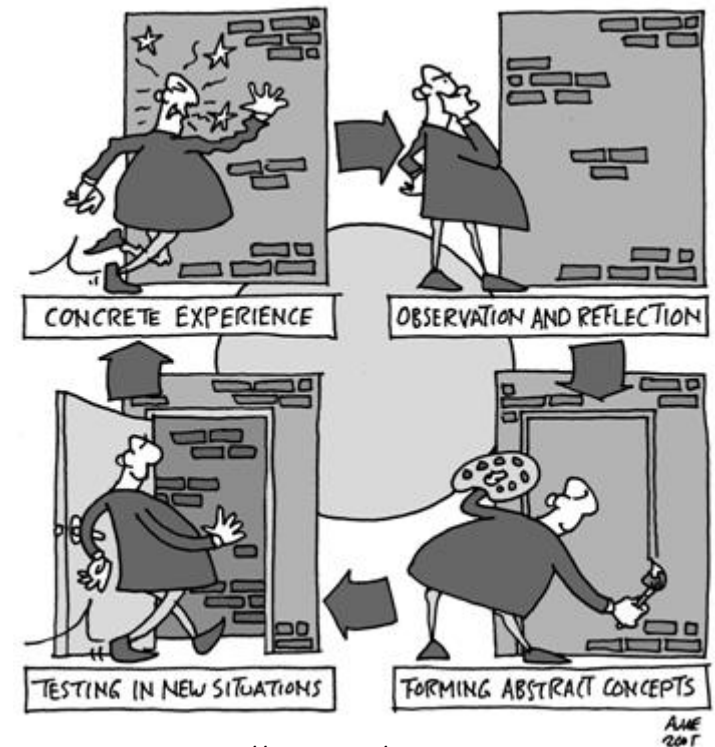
AR characteristics

- Based on **real life problems**
- Taking actions, aiming for change
- Knowledge creation (Learning) come from evaluation and reflection of the actions
- **Collaboration** with participants, democratic, interdisciplinary
- Researcher is immersed
- Situational, subjective
- Transferable rather than generalizable
- Interpretative, theory building

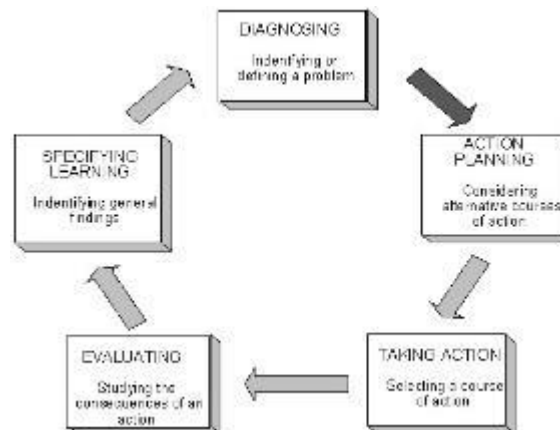
A cyclical approach



<https://edresearch.nmsu.edu/cro-research-publications-2/action-research-initiatives/ar/>

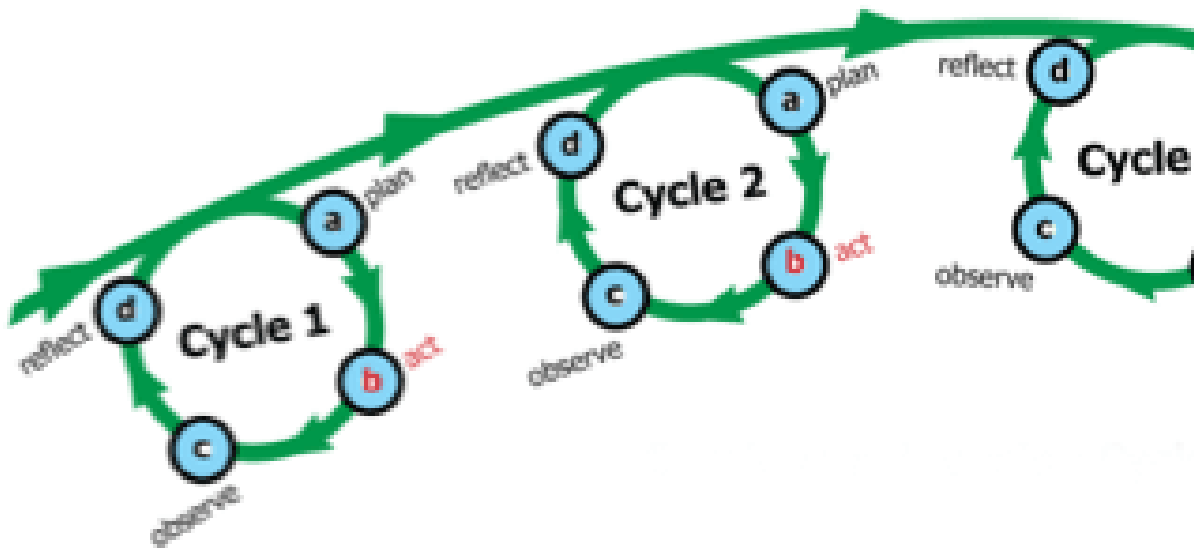


http://loop2.org/?page_id=175

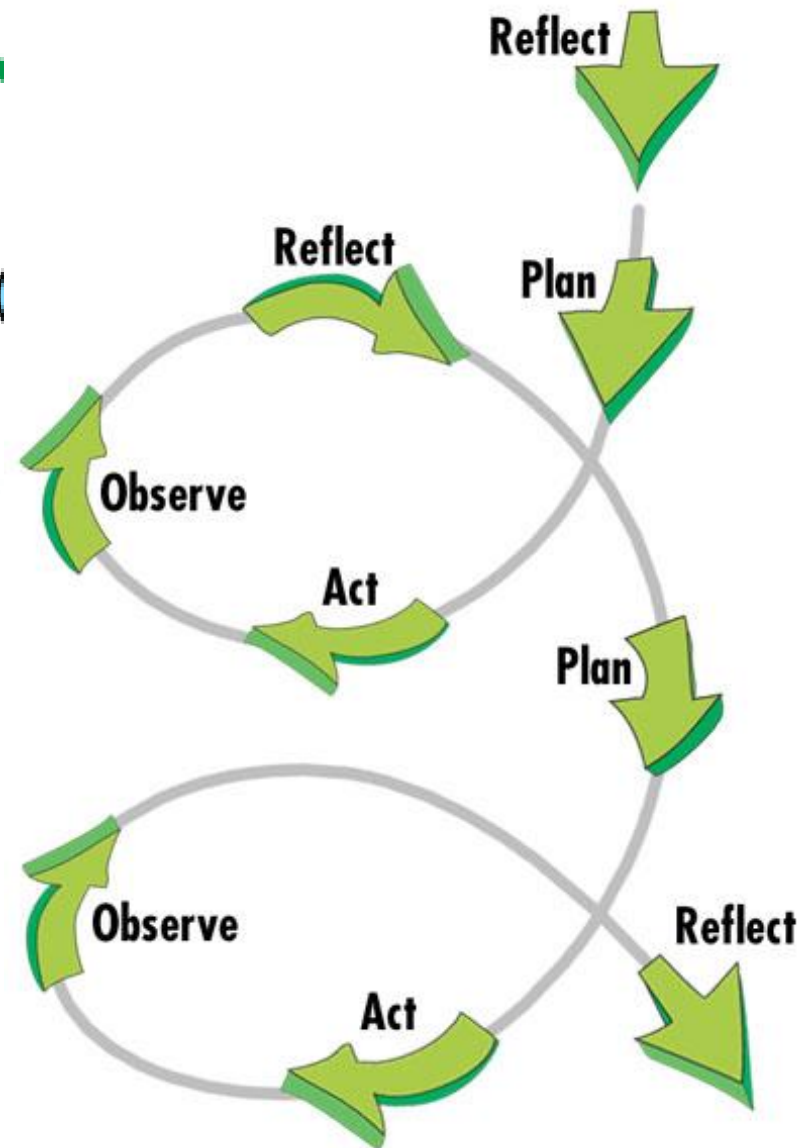


<http://www.vkmaheshwari.com/WP/?p=402>

An iterative approach



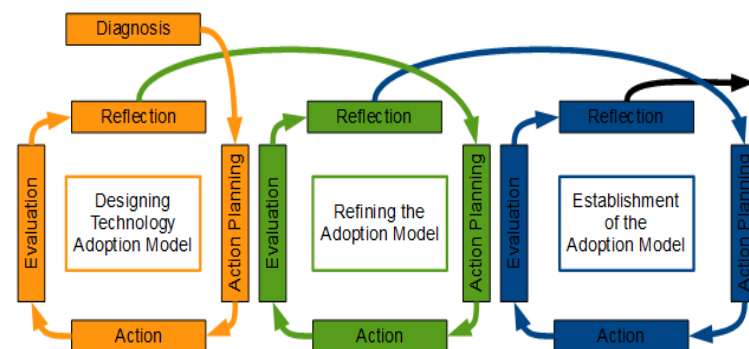
<http://robertsonandchang.com/services/learning-labs-action-research-for-innovation-and-excellence>



Developing a Technology adoption model for youth services in an informal settlement

Master Thesis by Asnath Kambunga

Cycle	Phase	Question/objective	Method	Finding/results
Cycle 1 Design Technology Adoption strategy	Diagnosis 1 (D1)	Which are the current technologies known to the Havana Youth? How are the youth currently adopting those technologies?	1. Questionnaire 2. Focus groups 3. Digital services expo workshop	Few technologies known, Yet no adoption, extracted factors affecting adoption in Havana
	Action Planning 1 (P1)	How is the rest of the world in similar setting as Havana adopting technologies?	1. Systematic literature review	Extracted factors affecting adoption Created strategy Created adoption time schedule
	Action 1 (A1)	How should new technologies be introduced in Havana to ensure better adoption?	1. Introduction to the adoption strategy workshop session 2. Training on the technology usage 3. Joint design of adoption campaign;	The campaign materials for deployment were not clear, youth were not committed
	Evaluation 1 (E1)	What were the action taken and their outcomes?	1. Synthesise results from AP1 and A1	Technology not adopted as the youth seems not to understand their roles and responsibilities, nor the system.
	Reflection 1 (R1)	What are the lesson learned to be considered in AP2	Reflect upon the action and its outcome. Summarise the lesson learned	Lesson learned to be carried over to AP2
Cycle 2 Refine Technology Adoption strategy	Action Planning 2 (AP2)	What are the elements that contribute to the ambassadors' understanding of their role in the technology adoption campaign?	Dedicated commitment session, Campaign material refinement Content training	Commitment session Campaigns materials redesign workshop, training plan
	Action 2 (A2)	How should the ambassadors understand their role in the technology adoption campaign to lead to better adoption of the technology?	1. Dedicated commitment session 2. Campaign materials refinement workshop 3. Content training	Youth set adoption/ deployment goals, better understanding of the system, partial execution of the adoption plan
	Evaluation 2 (E2)	What do the youth think of the whole adoption process?	Questionnaires	Youth not motivated
	Reflection 2 (R2)	What are the lesson learned to be considered in AP3	1. Summarise the lesson learned	Lesson learned to be carried over to AP3
Cycle 3 Establishment of the Technology Adoption strategy	Action Planning 3 (AP3)	What are the elements that will improve the technology adoption strategy	1. Scrutinize results from E2	Developed a refined adoption strategy, Developed a deployment schedule
	Action 3 (A3)	What are the best methods to implement the proposed adoption strategy in AP3?	1. Workshops 2. Meetings 3. Training	System deployed, motivated youth
	Evaluation 3 (E3)	How effective was the adoption strategy?	1. Questionnaires 2. Discussion among researchers	Committed ambassadors, Successful deployment
	Reflection 3 (R3)	What are the lesson learned to be considered in the development of a technology adoption model	1. Summarise the lesson learned	Social influence and capability plays a role in adoption



Kambunga, A. P., Winschiers-Theophilus, H., & Goagoses, N. (2018, December). Re-conceptualizing technology adoption in informal settlements based on a Namibian application. In *Proceedings of the Second African Conference for Human Computer Interaction: Thriving Communities* (pp. 1-10)

Data Collection Methods

- Semi-structured interviews
- Participant observations
- Analysis of documents
- Use of checklists: data registers, analysis tools, and health workers
- Software prototyping process and logs
- Group discussions and Training workshops
- Video/still pictures
- Analysis of press media reports

Data Analysis and Presentation

Interviews, Observations, Questionnaires, and site documents work together to support the research claims.

Empirical materials are presented in

- Descriptive statistics (quantitative data)
 - Model and measurement instrument for evaluating user satisfaction (7-point scale)
- Qualitative excerpts of encoded user reactions
- Software evaluation via criteria such as reliability and usability
- Logs and Screen shots of programs

Secondary Sources of Data

- Documents from the field
- Photos and videos

Validity of AR

Values

- Credibility
- Transferability
- Dependability
- Confirmability

Means

- Prolonged engagement, verbatim principle, multiple perspectives, workability
- Share context knowledge
- Transparency in description
- Present enough evidences

Hayes,G.R.2011.The relationship of action research to human-computer interaction. ACM Trans.Comput.Hum. Interact. 18, 3, Article 15 (July 2011), 20 pages.
DOI=10.1145/1993060.1993065 <http://doi.acm.org/10.1145/1993060.1993065>

Ethics in AR

- Conventional ones +
- Participants: from subject to collaborator
- Relationship
- Sustainability of change
- Choice of action

Design-based research

“The designing act of creating prototypes is in itself a potential generator of knowledge (if only its insights do not disappear into the prototype, but are fed back into the disciplinary and cross-disciplinary platforms that can fit these insights into the growth of theory).”

—(Stappers 2007)

<https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/research-through-design>

“we intend the term design research (sic) to mean an intention to produce knowledge and not the work to more immediately inform the development of a commercial product”. Zimmerman, Forlizzi, and Evenson (2007, p. 494)

RtD is defined as “a research approach that employs methods and processes from design practice [sic] as a legitimate method of inquiry” Zimmerman, Stolterman, and Forlizzi (2010)

Zimmerman, J., Forlizzi, J., & Evenson, S. (2007, April). Research through design as a method for interaction design research in HCI. In *Proceedings of the SIGCHI conference on Human factors in computing systems* (pp. 493-502)

Zimmerman, J., Stolterman, E., & Forlizzi, J. (2010, August). An analysis and critique of Research through Design: towards a formalization of a research approach. In *proceedings of the 8th ACM conference on designing interactive systems* (pp. 310-319)

Research through Design

“ the investigation of knowledge through purposeful design. ... the study of design and the process of knowledge production that occurs through the act of design”

Faste, T., & Faste, H. (2012, August). Demystifying “design research”: Design is not research, research is design. In *IDSA education symposium* (Vol. 2012, p. 15)

USER INTERFACE ACTIONS FOR THE RESTORATION OF EYE-TRACKED ATTENTION LOSS

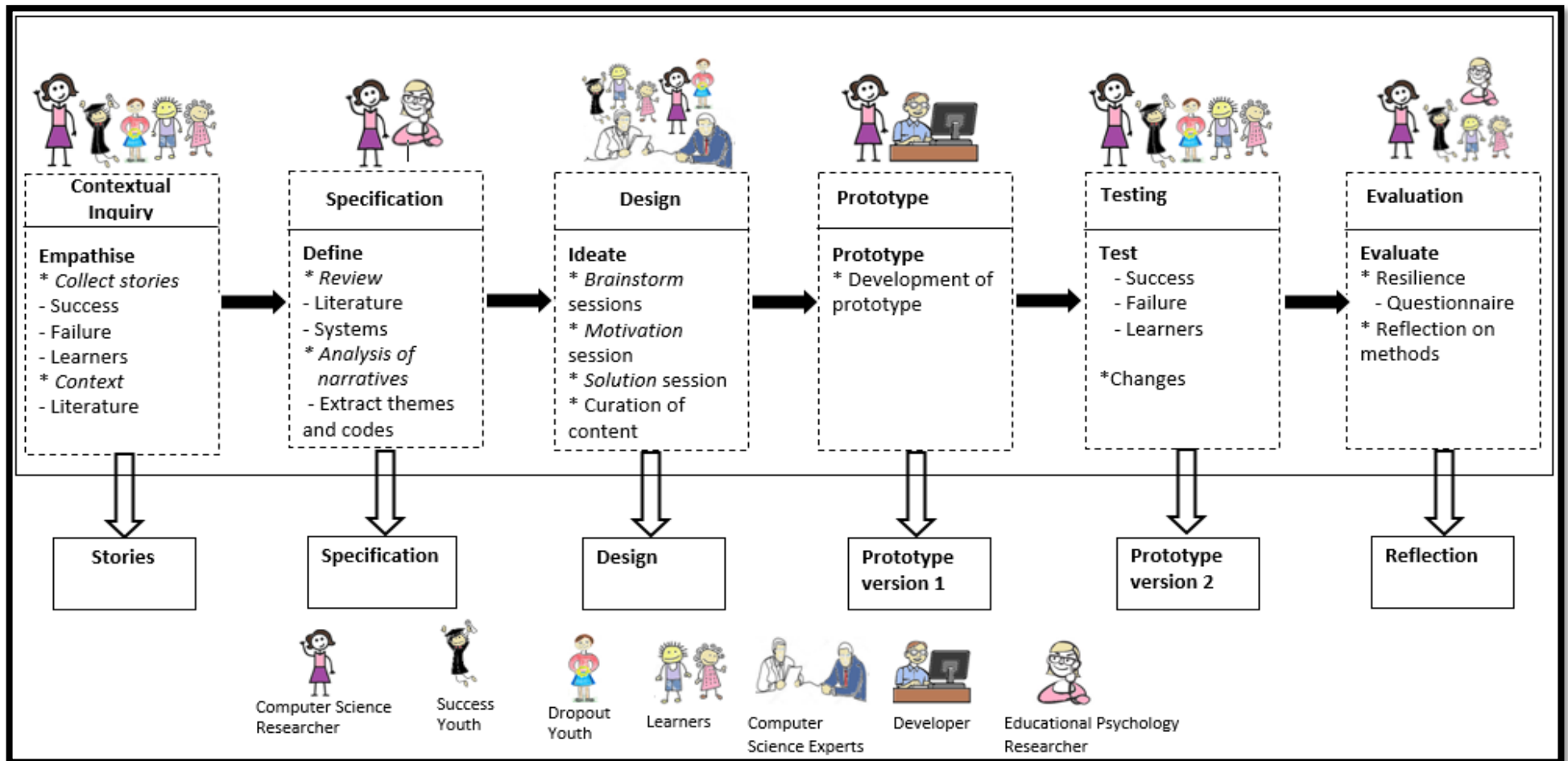
Honors thesis by Baptista Jacinto Joao

A Designed Experiment



RtD process

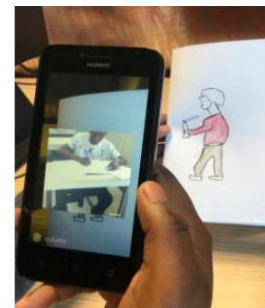
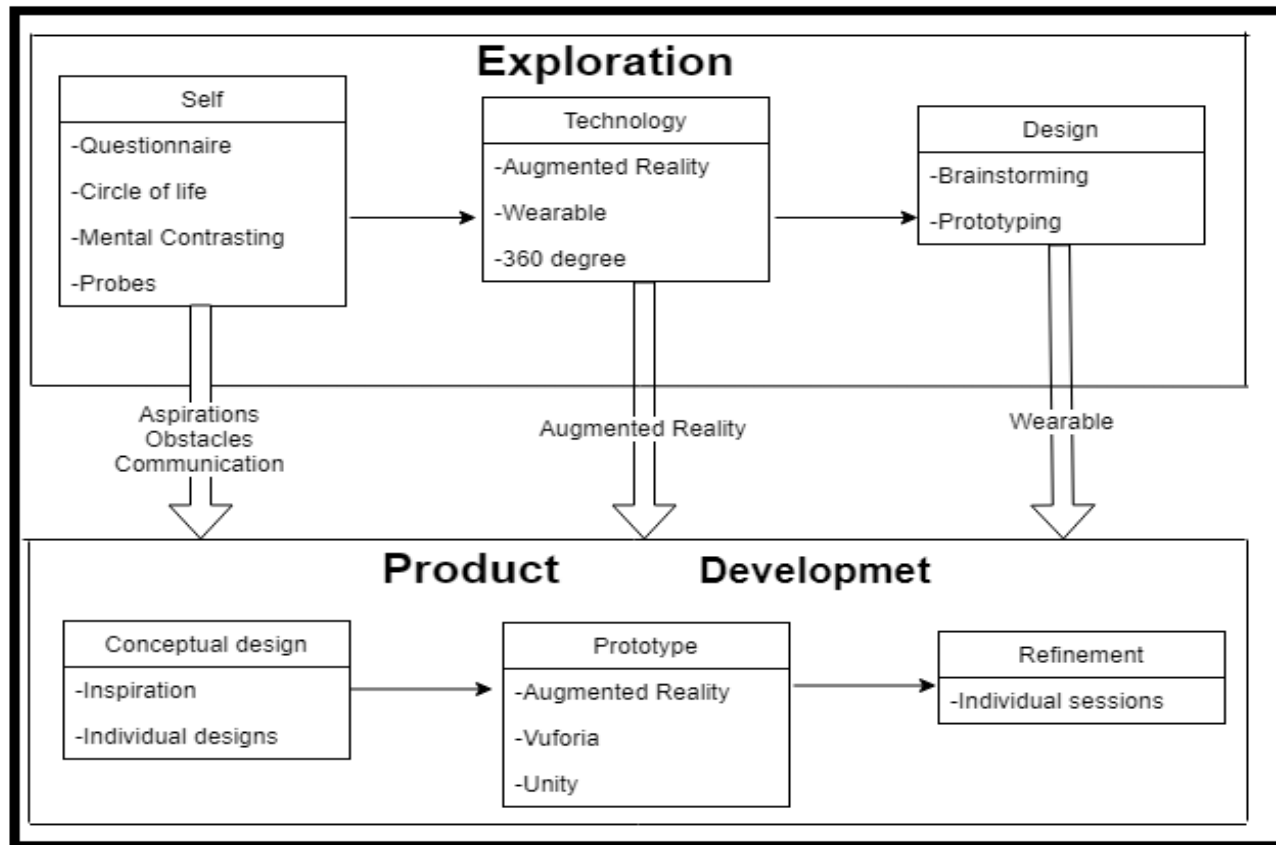
Technology to increase resilience to decrease school drop out



Kays, R. (2020) DEVELOPING TECHNOLOGY TO STRENGTHEN RESILIENCE IN SAN CHILDREN TO REDUCE SCHOOL DROPOUTS, Master Thesis

RtD process

Co-design of personal AR wearables to enhance social well-being



Kauhondamwa, M., Winschiers-Theophilus, M., Kapembe, S., Costa, H., Guxab, J., Kamati, I., & Afrikaner, H. (2018). Co-creating personal augmented reality accessories to enhance social well-being of urban San youth. *In Proceedings of the Second African Conference for Human Computer Interaction: Thriving Communities*



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